

CLEARWATER VALLEY IRRIGATION DISTRICT

ENGINEERING DIVISION

PRELIMINARY DESIGN MEMORANDUM

Surge/Holding Tank TK-2A

Cottonwood Creek Pumping Plant (PP-2)

To:

D. Harmon, P.E., District Engineer

From:

T. Frederickson, P.E., Facilities Engineer

Date:

July 18, 2025

CVID Project No.:

CIP-2025-017

Subject:

Preliminary Design Parameters and Geotechnical Investigation
Scope for Proposed Surge/Holding Tank TK-2A

Reference Documents:

1. WGC Report No. 2018-GEO-0093 (October 2018)

2. WGC Field Inspection Memo — Foundation Excavation (March 2019)

3. CVID 2024 Annual Facility Condition Assessment — PP-2

4. CVID Hydraulic Analysis, Phase 2 System Improvements (April 2025)

1.0 PURPOSE

This memorandum presents the preliminary design parameters for the proposed surge/holding tank (designated TK-2A) at the Cottonwood Creek Pumping Plant (PP-2) and defines the scope of the supplemental geotechnical investigation required to support final design. The purpose is to provide the geotechnical consultant with sufficient project information to prepare a focused and cost-effective investigation scope.

2.0 BACKGROUND AND PROJECT NEED

The Clearwater Valley Regional Water Conveyance System delivers irrigation and municipal water from Cottonwood Creek to users throughout the district. PP-2 was commissioned in June 2019 as part of Phase 1 and has operated reliably for five irrigation seasons. The Phase 2 Hydraulic Analysis (April 2025) identified a need for surge protection and operational storage at the PP-2 site to address the following conditions:

- **Hydraulic transients:** Simultaneous startup or shutdown of multiple pumps generates pressure surges in the 36-inch transmission main. The current system relies on pump sequencing controls and pressure relief valves, which are adequate for normal operations but provide limited protection during power outage or emergency shutdown scenarios.
- **Operational flexibility:** A holding tank would allow pumps to operate at optimal efficiency (full speed) while decoupling pumping rates from instantaneous downstream demand. This is projected to reduce energy costs by 8 to 12 percent and extend pump bearing life by reducing start/stop cycles.

- **System redundancy:** The tank would provide approximately 2 hours of storage at average demand rates, allowing continued water delivery during short-term pump maintenance or power interruptions.

3.0 PROPOSED TANK DESIGN PARAMETERS

The following preliminary design parameters have been established based on the hydraulic analysis and district operational requirements:

Table 1: Preliminary Design Parameters — Tank TK-2A

Parameter	Value	Notes
Tank type	Welded steel, flat-bottom, ground-supported	AWWA D100 design standard
Nominal capacity	250,000 gallons	2 hrs at avg. demand
Diameter	40 ft	Preliminary; final per manufacturer
Shell height	28 ft	Includes 2 ft freeboard
Operating water depth	26 ft max.	Overflow at 26 ft
Foundation type	Ringwall (concrete)	Per AWWA D100 Sec. 14
Ringwall diameter	~42 ft (outside)	2 ft wider than shell
Ringwall depth	4 ft below grade (min.)	Below frost/swell zone
Tank floor	Steel on compacted fill pad	6-inch sand bedding
Max. foundation load	~4,500 psf (shell bearing)	At ringwall, full tank
Max. uniform load	~1,800 psf (interior floor)	Water weight on pad
Seismic importance	Category III	Essential facility
Design wind speed	115 mph (3-sec gust)	ASCE 7-22, Risk Cat. III

4.0 PROPOSED SITE AND LOCATION

The proposed tank location is approximately 120 feet southeast of the existing PP-2 pump bay, on the same alluvial terrace surface. The site is currently undeveloped with native grass and sagebrush cover. The ground surface elevation at the proposed tank center is approximately 5,482 ft (NAVD88), which is approximately 2 to 3 feet lower than the PP-2 pump bay elevation (5,484.7 ft at BH-PP2-01). This slight elevation difference is attributed to the natural southward slope of the terrace toward the valley floor.

The proposed location was selected based on the following criteria: (a) proximity to the existing discharge manifold for hydraulic connection; (b) adequate separation from the existing structures for construction access; (c) location on the same geomorphic surface as the existing facility (same expected subsurface conditions); and (d) avoidance of the CVID canal right-of-way and existing utility easements.

The nearest existing boring (BH-PP2-04) is approximately 100 feet northwest of the proposed tank center. Based on the consistency of subsurface conditions observed across four borings spanning approximately 180 feet at the PP-2 site during the 2018 investigation, and confirmed during 2019 construction, the geologic conditions at the proposed tank location are expected to be consistent with the established subsurface model.

5.0 EXISTING GEOTECHNICAL DATA

The following geotechnical data are available from the original PP-2 investigation and subsequent construction and monitoring:

WGC Report No. 2018-GEO-0093 (October 2018): Four borings (BH-PP2-01 through BH-PP2-04), 25 to 40 ft deep, with comprehensive laboratory testing including Atterberg limits, gradation with hydrometer, swell/consolidation, corrosion, UCS (12 tests), and ITS (14 tests). The report establishes a four-layer subsurface model: (1) alluvial fill, 2 to 4 ft; (2) lean clay (CL), 6 to 10 ft thick; (3) alluvial sand/silt (SM/ML), extending to bedrock; (4) Challis Volcanic sandstone/siltstone bedrock at 20.5 to 23 ft. Groundwater at 13.5 to 14.2 ft bgs.

WGC Field Inspection Memo (March 2019): Confirmed subsurface conditions during construction were consistent with the investigation. Six drilled shafts installed with bedrock at 21.0 to 22.5 ft. No unexpected conditions encountered.

CVID 2024 Facility Assessment: Five years of foundation performance data confirming settlement within predictions (max 0.22 inches on spread footings, negligible on drilled shafts), swell mitigation performing as designed, no corrosion distress. Groundwater measured at 15.8 to 16.1 ft bgs in August 2024.

6.0 GEOTECHNICAL DESIGN QUESTIONS

The supplemental geotechnical investigation must address the following design questions specific to the TK-2A tank foundation:

6.1 Confirmation of Subsurface Model

Do subsurface conditions at the proposed tank location match the four-layer model established by the 2018 investigation? Specifically, what is the thickness of the clay stratum and the depth to bedrock at the tank location?

6.2 Ringwall Foundation Bearing Capacity

Is the allowable bearing pressure of 3,000 psf established in the 2018 report adequate for the ringwall foundation load of approximately 4,500 psf? If not, what foundation modifications are required (deeper embedment, wider footing, drilled shaft support)?

6.3 Settlement Under Tank Loads

What is the estimated total and differential settlement under the tank loading conditions? AWWA D100 limits differential settlement to 1/2 inch across the tank diameter (40 ft). Given the overconsolidated clay, is the tank loading expected to exceed the preconsolidation pressure?

6.4 Swell Potential and Mitigation

What swell mitigation measures are required for the ringwall and tank floor pad? Is the 2-foot over-excavation and replacement approach used for PP-2 adequate, or does the circular geometry of the tank require a different strategy?

6.5 Corrosion Environment

Are corrosion conditions at the tank location consistent with those at PP-2? The steel tank shell will extend below grade at the ringwall; is external coating/cathodic protection required beyond the measures specified in

the 2018 report?

6.6 Bedrock Characterization

What is the strength and quality of the bedrock at the tank location? If the bearing capacity analysis requires deeper foundations (drilled shafts to bedrock), what are the applicable design parameters?

6.7 Seismic Performance

Are the seismic site classification and design parameters from the 2018 report applicable to the tank location? Is the tank foundation susceptible to seismically induced settlement or lateral spreading?

7.0 RECOMMENDED SUPPLEMENTAL INVESTIGATION SCOPE

Based on the available geotechnical data, the consistency of conditions documented across the PP-2 site, and the five years of satisfactory foundation performance, a focused supplemental investigation of limited scope is recommended. The investigation should include the following:

7.1 Field Exploration

One exploratory boring (BH-TK2A-01) at the approximate center of the proposed tank footprint, advanced to a minimum depth of 30 feet below existing grade (or 10 feet into competent bedrock, whichever is deeper). The boring should include:

- Hollow-stem auger drilling with SPT sampling at 2.5-foot intervals through the alluvial overburden (consistent with 2018 methods for direct comparison);
- Shelby tube sampling of the clay stratum for swell/consolidation testing (one sample from the mid-depth of the clay layer);
- HQ wireline diamond coring of the bedrock (minimum two 5-foot core runs) for UCS and ITS testing;
- Installation of a groundwater monitoring standpipe; and
- Field measurement of groundwater level 24 hours after drilling.

7.2 Laboratory Testing

The laboratory testing program should be designed to verify that conditions at the tank location are consistent with the 2018 data and to fill any data gaps specific to the tank foundation design. Recommended testing includes:

Table 2: Recommended Laboratory Testing Program

Test	Standard	No. of Tests	Purpose
Atterberg limits	ASTM D4318	2	Verify clay classification matches 2018 data
Gradation w/ hydrometer	ASTM D6913/D7928	1	Confirm particle size distribution
Swell/consolidation	ASTM D4546	1	Swell potential & preconsolidation at tank site
Corrosion suite	ASTM G57, D4972, D4327	2	Verify corrosion class (clay and sand)
UCS with strain	ASTM D7012-C	1-2	Bedrock strength, modulus, Poisson's ratio
Indirect tensile (ITS)	ASTM D3967	8-10	Tensile strength variability, statistical basis
Moisture content	ASTM D2216	4-6	Index property across all layers

Test	Standard	No. of Tests	Purpose
Core logging / RQD	ASTM D6032	All core	Rock quality assessment

7.3 Engineering Deliverables

The geotechnical consultant's report should address each of the design questions listed in Section 6.0 and provide the following:

- Comparison of the 2026 supplemental data to the 2018 investigation results, with a specific assessment of whether the existing geotechnical model remains valid for the tank location;
- Foundation recommendations for the ringwall and tank floor pad, including allowable bearing pressures, embedment depth, and fill specifications;
- Settlement estimates under the full-tank loading condition, including assessment of differential settlement across the 40-foot tank diameter;
- Swell mitigation recommendations specific to the circular ringwall geometry;
- Corrosion protection recommendations for the below-grade steel shell and embedded anchor bolts; and
- Seismic design parameters and assessment of liquefaction potential.

8.0 JUSTIFICATION FOR SINGLE-BORING SCOPE

A single supplemental boring is considered adequate for the proposed tank foundation based on the following factors:

1. **Geologic Consistency:** The original 2018 investigation included four borings spanning approximately 180 feet across the PP-2 site. All four borings encountered the same four-layer stratigraphic sequence with consistent layer thicknesses, material properties, and groundwater conditions. The proposed tank location is on the same geomorphic surface (alluvial terrace) approximately 120 feet from the nearest existing boring.
2. **Construction Verification:** The 2019 construction inspection confirmed that subsurface conditions exposed during excavation and shaft installation were consistent with the borings at all six drilled shaft locations. No unexpected conditions were encountered anywhere on the site.
3. **Performance Record:** Five years of foundation performance monitoring confirm that the geotechnical model accurately represents site conditions. Settlement, swell, and corrosion performance are all within predicted ranges.
4. **Geologic Setting:** The alluvial terrace is a uniform depositional environment with laterally continuous strata. The underlying bedrock is a flat-lying, laterally persistent sedimentary sequence. There are no known geologic features (faults, paleochannels, karst) that would cause abrupt lateral changes in subsurface conditions within the site area.
5. **Industry Practice:** The supplemental investigation follows accepted practice for adjacent-facility additions where comprehensive geotechnical data already exist. The 2024 Facility Assessment specifically recommended this approach (Section 5.3).

If the supplemental boring reveals conditions that deviate significantly from the established subsurface model (for example, unexpected soft zones, absent clay stratum, or substantially different bedrock depth), additional

borings will be required and the investigation scope will be expanded accordingly.

9.0 SCHEDULE AND BUDGET

The supplemental geotechnical investigation is planned for January 2026. Drilling during the winter (non-irrigation) season avoids conflicts with district operations and provides a groundwater measurement during a period comparable to the original 2018 investigation (September measurements). The estimated budget for the supplemental investigation, including drilling, laboratory testing, and engineering report, is \$18,000 to \$24,000. This compares favorably to the \$65,000 cost of the original four-boring investigation, reflecting the reduced scope made possible by the existing data.

The geotechnical report is needed by April 2026 to support the 60-percent design milestone. Tank procurement (long-lead item) is scheduled for July 2026, with construction anticipated in the fall 2026 non-irrigation season.

Prepared by:

T. Frederickson, P.E.

Facilities Engineer, CVID

Idaho PE-11247

Approved by:

D. Harmon, P.E.

District Engineer, CVID

Idaho PE-8834

cc: M. Salazar, Operations Superintendent; CVID Engineering Files; Western Geotechnical Consultants (under separate cover with investigation authorization)